Naive Bayes

Prerequisites: Basics of probability and Bayes theorem

Objectives:

- Understand the basics of probability-based learning
- Fundamentals of Bayes theorem
- Introduction to Bayes classification problem
- Introduction to the Naïve Bayes classification problem based on Bayes theorem
- Explore the Naïve Bayes classification algorithm

What is Probability-based learning?

- Here, learning combines **prior knowledge or prior probabilities** with observed data
- Instead of giving a single, fixed answer, this method calculates the **likelihood of different possible results.**
- Uses the concept of **probabilistic theory** that **describes randomness, uncertainty, and noise to predict the future events**
- Tool for modeling large datasets

Marginal, joint, and conditional probability

- Joint probability is the probability of two events occurring simultaneously.
- Marginal probability is the probability of an event irrespective of the outcome of another variable.
- Conditional probability is the probability of one event occurring in the presence of a second event.

Example to understand Marginal, joint, and conditional probability

Probability table

Gender	Pass	Fail	Total
Male	40	10	50
Female	30	20	50
Total	70	30	100

Joint Probability (Probability of Male AND Pass) :P(Male∩Pass)=40/100 =0.40
So Joint probability of Male=40%

Similarly (Female∩Pass)=30/100=0.30

Probability of being Male:P(Male)=50/100 =0.50
So **Marginal probability of Male**=50% ,
Marginal probability of passing= P(P)=70/100= 70%

Since we already know the student is male, we consider only the 50 males. 40 of them passed:
So **conditional probability is P(Pass|Male)=**40/50=0.80 = 80%

Important relationship on Marginal, joint, and conditional probability

Probability table

Gender	Pass	Fail	Total
Male	40	10	50
Female	30	20	50
Total	70	30	100

Joint Probability (Probability of Male AND Pass) :$P(M \cap P) = P(P|M) \times P(M)$
[i.e. product of conditional probability & Marginal Probability]
Joint Probability is Symmetrical : $P(M \cap P) = P(P \cap M)$
Conditional Probability is not Symmetrical => $P(P|M) \neq P(M|P)$

Conditional Probability (Probability of Male AND Pass) :$P(P|M) = P(M \cap P) / P(M)$

conditional probability is $P(P|M) = 40/50 = 0.80 = 80\%$

An Alternate Way To Calculate Conditional Probability

- Specifically, one conditional probability can be calculated using the other conditional probability;
- P(A|B) = P(B|A) * P(A) / P(B)
- The reverse is also true;
- P(B|A) = P(A|B) * P(B) / P(A)
  - This alternate approach of calculating the conditional probability is useful either **when the joint probability is challenging to calculate** (which is most of the time), or
  - when the reverse conditional probability is available or easy to calculate.
- This **alternate calculation of the conditional probability is referred to as Bayes Rule or Bayes Theorem**, named for Reverend Thomas Bayes, who is credited with first describing it.

What is Prior Probability?

- Class priors refer to the prior probabilities of each class in a classification problem.
- They represent the probability of a class occurring before any evidence (features) is observed.

- The class prior P(C) for a class C is defined as:

$$P(C) = \frac{\text{Number of samples in class } C}{\text{Total number of samples}}$$

- Consider a dataset where: 100 samples belong to Class A, 50 samples belong to Class B.

$$P(\text{Class A}) = \frac{100}{150} = 0.67$$

$$P(\text{Class B}) = \frac{50}{150} = 0.33$$

What is Conditional Probability?

- Conditional probability is the **probability of an event occurring, given that another event has already occurred**
- It measures the likelihood of one event under the assumption that we know another event has occurred
- The conditional probability of event A, given that event B has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- Where:
- P(A|B): The probability of A occurring, given B has occurred.
- $P(A \cap B)$: The probability of both A and B occurring (joint probability).
- P(B): The probability of B occurring (prior probability, must be >0).

Conditional Probability

- For example:

➢Consider 2 events A&B: “It Rains”, “It is cloudy”

➢Assuming P(A)=0.3(Rain on any given day is 30% likely).

&, P(B)=0.5(Cloudiness is 50% likely).

$P(A \cap B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(A \cap B)$ (A∩B)=0.2 (Rain and cloudiness together occur 20% of the time).

e conditional probability of rain, given that it is cloudy, is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = 0.4$$

What is Posterior Probability?

- is the probability of an event or hypothesis A occurring after observing some evidence B.

posterior probability is given by Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Bayesian Learning

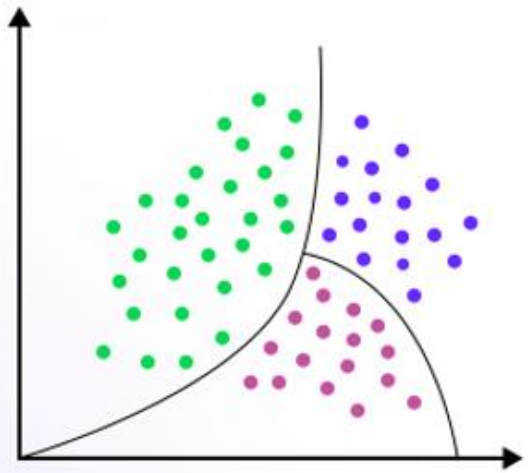
- Bayesian learning describes and represents knowledge in an uncertain domain
- It mainly uses the probability measure to reason about this knowledge
- Uses Bayes theorem to infer the unknown parameters
- Useful applications of Bayesian learning are in reasoning and diagnosis
- It’s useful in handling missing data and estimating uncertainty in prediction

What is Bayes Theorem?

- Is a fundamental concept in probability theory
- Plays a crucial role in various machine learning algorithms
- Provides a way to update the probability based on new evidence or information
- Allows us to infer our belief in a hypothesis based on new evidence
- When we use Bayes theorem we use our prior belief in the hypothesis represented by P(A)

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Understanding Bayes' Theorem



The formula and its components

- P(A|B) is the posterior probability - the probability of class A given feature B
- P(B|A) is the likelihood - the probability of feature B given class A
- P(A) is the prior probability of class A
- P(B) is the evidence - the probability of feature B

At its heart, Bayes' theorem provides a way to calculate conditional probability—the likelihood of an event occurring based on prior knowledge of conditions related to the event. The formula looks like this:

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

Breaking this down:

- P(A|B) is the posterior probability – the probability of class A given feature B
- P(B|A) is the likelihood – the probability of feature B given class A
- P(A) is the prior probability of class A
- P(B) is the evidence – the probability of feature B

Prior, likelihood, and posterior explained

These components work together in a logical sequence:

1. **Prior** probability represents your initial belief about how likely an outcome is before seeing any evidence. For instance, if 30% of emails in your dataset are spam, the prior probability of any new email being spam is 0.3.
2. **Likelihood** measures how probable the observed evidence is, given a particular outcome. In spam detection, this might be the probability of seeing the word “discount” in spam emails versus non-spam emails.
3. **Posterior** probability is your updated belief after considering the evidence. It represents the final probability that helps you make a decision. This is what the Naive Bayes classifier calculates to determine which class a data point belongs to.

How Bayes' Theorem applies to classification?

In machine learning classification, Bayes' theorem helps determine which category a data point belongs to based on its features. Here's how it works:

- Start with prior probabilities for each class based on training data
- Calculate the likelihood of observing specific features in each class
- For a new data point, compute the posterior probability for each possible class
- Assign the data point to the class with the highest posterior probability

Bayes Theorem

- If we have P(A), then we can calculate P(not A) as its complement;
- P(not A) = 1 – P(A)
- Additionally, if we have P(not B|not A), then we can calculate P(B|not A) as its complement;
- P(B|not A) = 1 – P(not B|not A)

Naive Bayes

- Naive Bayes is a simple and powerful algorithm for predictive modeling.
- Naïve Bayes algorithm is a supervised learning algorithm, which is based on **Bayes theorem** and used for solving classification problems.

Naive Bayes Classifier

- Naive Bayes is a *classification algorithm for binary (two-class) and multiclass classification problems*.
- The technique is *easiest to understand when described using binary or categorical input values*.
- It is called **naive Bayes or idiot Bayes** because *the calculation of the probabilities for each hypothesis are simplified to make their calculation tractable*.
- Rather than attempting *to calculate the values of each attribute value P(d1, d2, d3|h), they are assumed to be conditionally independent given the target value and calculated as P(d1|h) x P(d2|h) and so on*.
- This is a very strong assumption that is most unlikely in real data, i.e. that the attributes do not interact.
- Nevertheless, the approach performs surprisingly well on data where this assumption does not hold.

Why is this called “naive”??

- Naive Bayes assumes that all features are independent of each other when the class is known.
- For example, suppose we want to predict whether a person will Play based on:
  - Weather = Sunny
  - Temperature = Mild
  - Humidity = High
- Naive Bayes assumes:
  - Once we know whether the person will Play or Not Play, Weather, Temperature, and Humidity are independent of each other.

- Classifying these Naive features using Bayes theorem is known as Naive Bayes.
- Counting how many times each attribute co-occurs with each class is the main learning idea for Naive Bayes classifier.

Working of Naïve Bayes Classifier –Example 1

Working of Naïve Bayes Classifier -Steps

1. Convert the given dataset into frequency tables.
2. Generate Likelihood table by finding the probabilities of given features.
3. Use Bayes theorem to calculate the posterior probability.
4. Check which class has a higher probability, given the input belongs to the higher probability class.

Working of Naïve Bayes Classifier -Example

- Suppose we have a dataset of weather conditions and corresponding target variable "Play".
- Using this dataset, decide whether player should play or not on a particular day according to the weather conditions.
- Problem: If the weather is sunny, then the Player should play or not?
- P(yes|sunny) > p(no|sunny) ?
- consider the dataset:

	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Working of Naïve Bayes Classifier -Example

- Step 1: Convert the given dataset into frequency tables

weather	yes	no
Overcast		
rainy		
sunny		
total		

- Step 2: Convert into Likelihood table

weather	yes	no	probability
Overcast			
rainy			
sunny			
All probability			

Working of Naïve Bayes Classifier -Example

- Step 1 & 2: Convert the given dataset into frequency tables

- Convert into Likelihood table

weather	yes	no	probability
Overcast	5	0	5/14=0.357
rainy	2	2	4/14=0.29
sunny	3	2	5/14=0.357
Total	10	4	
All probability	10/14= 0.71	4/14=0.29	

Working of Naïve Bayes Classifier –Example
Step 1&2: Convert the given dataset into frequency tables and Likelihood table

Frequency table for the Weather Conditions:

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	5

Likelihood table weather condition:

Weather	No	Yes	
Overcast	0	5	5/14= 0.35
Rainy	2	2	4/14=0.29
Sunny	2	3	5/14=0.35
All	4/14=0.29	10/14=0.71	

Note: In frequency table, total of “No”= 4

Working of Naïve Bayes Classifier –Example
Step3: **Applying Bayes' theorem**

- Problem: If the weather is sunny, then the Player should play or not?
- **P(Yes|Sunny)= P(Sunny|Yes)*P(Yes)/P(Sunny)**

- P(Sunny|Yes)= 3/10= 0.3

- P(Sunny)= 5/14= 0.35

- P(Yes)=10/14=0.71

- So P(Yes|Sunny) = 0.3*0.71/0.35= 0.60

Working of Naïve Bayes Classifier –Example
Step3: **Applying Bayes' theorem**

- Problem: If the weather is sunny, then the Player should play or not?
- **P(No|Sunny)= P(Sunny|No)*P(No)/P(Sunny)**
- **P(Sunny|No)= 2/4=0.5**
- **P(No)= 4/14=0.29**
- **P(Sunny)= 5/14=0.35**
- **So P(No|Sunny)= 0.5*0.29/0.35 = 0.41**
- **So as we can see from the above calculation that P(Yes|Sunny)>P(No|Sunny)**
- **Hence on a Sunny day, Player can play the game.**

Working of Naïve Bayes Classifier – Example 2

Working of Naïve Bayes Classifier - Example

- Data of student's effort level(Poor, Average and Good) and their results(Pass and Fail)

EFFORT	RESULT
Poor	Fail
Average	Pass
Average	Pass
Good	Pass
Good	Pass
Poor	Fail
Poor	Fail
Poor	Pass
Poor	Fail
Average	Pass
Average	Fail
Good	Pass
Average	Fail

Check if this statement is correct. “Student will fail if the efforts are poor”.

Working of Naïve Bayes Classifier -Example

EFFORT	RESULT
Poor	Fail
Average	Pass
Average	Pass
Good	Pass
Good	Pass
Poor	Fail
Poor	Fail
Poor	Pass
Poor	Fail
Average	Pass
Average	Fail
Good	Pass
Average	Fail

Check if this statement is correct. “Student will fail if the efforts are poor”.

We can use Bayes Theorem as,

P(Fail | Poor) = P(Poor | Fail) * P(Fail) / P(Poor)

Working of Naïve Bayes Classifier -Example

- “Student will fail if the efforts are poor”: P(Fail | Poor)
- P(Poor | Fail) = Number of students who failed with poor efforts / Number of students failed = 4 / 6 = 0.66

- P(Fail) = Number of students failed / Total students = 6 / 13
- P(Poor) = Number of students with poor efforts / Total students = 5 / 13

- P(Fail | Poor) = 0.66 * (6/13) / (5/13) = 0.66 * 6 / 5 = 0.792

- This is a higher probability for “Student will fail if the efforts are poor”.

BAYES' THEOREM

- Bayes' theorem describes the probability of occurrence of an event related to any condition. It is also considered for the case of conditional probability.
- Bayes theorem is also known as the formula for the probability of "causes"

- **Bayes Theorem Derivation** ' theorem can be derived for events and random variables separately using the definition
- Bayes of conditional probability and density.

From the definition of conditional probability, Bayes theorem can be derived for events as given below:

$P(A|B) = P(A \cap B)/ P(B)$, where $P(B) \neq 0$

$P(B|A) = P(B \cap A)/ P(A)$, where $P(A) \neq 0$

Here, the joint probability $P(A \cap B)$ of both events A and B being true such that,

$P(B \cap A) = P(A \cap B)$

$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A)$

$P(A|B) = [P(B|A)\, P(A)]/ P(B)$, where $P(B) \neq 0$

Terms in BAYES' THEOREM

- The terms in the Bayes Theorem equation are given names depending on the context where the equation is used. It can be helpful to think about the calculation from these different perspectives and help to map your problem onto the equation.
- Firstly, in general, the result P(A|B) is referred to as the posterior probability and P(A) is referred to as the prior probability.

P(A|B): Posterior probability.

P(A): Prior probability.

Sometimes P(B|A) is referred to as the likelihood and P(B) is referred to as the evidence.

P(B|A): Likelihood.

P(B): Evidence.

This allows Bayes Theorem to be restated as:

Posterior = Likelihood * Prior / Evidence

Example Problem on Bayes theorem

Suppose the probability of the weather being cloudy is 40%. Also suppose the probability of rain on a given day is 20%. Also suppose the probability of clouds on a rainy day is 85%. If it’s cloudy outside on a given day, what is the probability that it will rain that day?

NAÏVE BAYES CLASSIFIER

- The Naive Bayes classifier is a simple yet powerful machine learning algorithm that is particularly well-suited for tasks involving text classification and spam filtering.
- It is based on Bayes' theorem and makes the "naive" assumption of independence between the features used for classification

Key Concepts of Naive Bayes Classifier

- 1. Bayes' Theorem: At the heart of the Naive Bayes classifier is Bayes' theorem, a fundamental concept in probability theory. It allows us to calculate the probability of a particular event based on prior knowledge of conditions that might be related to the event.
- Bayes' theorem can be expressed as: $$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Where:

- P(A|B) is the conditional probability of event A given event B.
- P(B|A) is the conditional probability of event B given event A.
- P(A) and P(B) are the probabilities of events A and B, respectively.

Key Concepts of Naive Bayes Classifier(contd..)

- 2. Independence Assumption: The "naive" part of Naive Bayes comes from assuming that the **features used for classification are independent of each other**. In practice, this assumption is often not entirely accurate, but the model can still perform well.
- 3. **Multinomial and Gaussian Naive Bayes**: There are different variants of Naive Bayes classifiers, including Multinomial and Gaussian Naive Bayes.
- Multinomial Naive Bayes is commonly used for text data, while Gaussian Naive Bayes is suitable for continuous data with a normal distribution

Naïve Bayes with multiple features

Naïve Bayes Multi-class classification model

Naïve Bayes with multiple features - Example

Calculate the probability of playing when the weather is overcast and the temperature is mild.

Whether	Temperature	Play
Sunny	Hot	No
Sunny	Hot	No
Overcast	Hot	Yes
Rainy	Mild	Yes
Rainy	Cool	Yes
Rainy	Cool	No
Overcast	Cool	Yes
Sunny	Mild	No
Sunny	Cool	Yes
Rainy	Mild	Yes
Sunny	Mild	Yes
Overcast	Mild	Yes
Overcast	Hot	Yes
Rainy	Mild	No

Naïve Bayes with multiple features - Example

1. Calculate **prior probability** for given class labels
2. Calculate **conditional probability with each attribute** for each class
3. **Multiply same class conditional** probability
4. Multiply prior probability with step 3 probability
5. Check the class with higher probability. Higher probability class belongs to given test data

Naïve Bayes with multiple features - Example

- Calculate the probability of playing when the weather is overcast, and the temperature is mild.
- **Probability of playing**

- P(Play= Yes | Weather=Overcast, Temp=Mild) = P(Weather=Overcast, Temp=Mild | Play= Yes)P(Play=Yes) .....(1)
- P(Weather=Overcast, Temp=Mild | Play= Yes)= P(Overcast |Yes) P(Mild |Yes) …….(2)
- Calculate Prior Probabilities: P(Yes)= 9/14 = 0.64
- Calculate Posterior Probabilities: P(Overcast |Yes) = 4/9 = 0.44
- P(Mild |Yes) = 4/9 = 0.44
- P(Weather=Overcast, Temp=Mild | Play= Yes) = 0.44 * 0.44 = 0.1936(Higher)
- P(Play= Yes | Weather=Overcast, Temp=Mild) = 0.1936*0.64 = 0.124

Naïve Bayes with multiple features - Example

- Calculate the probability of playing when the weather is overcast, and the temperature is mild.
- **Probability of not playing:**

- P(Play= No | Weather=Overcast, Temp=Mild) = P(Weather=Overcast, Temp=Mild | Play= No)P(Play=No) .....(3)
- P(Weather=Overcast, Temp=Mild | Play= No)= P(Weather=Overcast |Play=No) P(Temp=Mild | Play=No) ……..(4)
- Calculate Prior Probabilities: P(No)= 5/14 = 0.36
- Calculate Posterior Probabilities: P(Weather=Overcast |Play=No) = 0/9 = 0
- P(Temp=Mild | Play=No)=2/5=0.4
- P(Weather=Overcast, Temp=Mild | Play= No) = 0 * 0.4= 0
- P(Play= No | Weather=Overcast, Temp=Mild) = 0*0.36=0
- The probability of a 'Yes' class is higher. So you can say here that if the weather is overcast than players will play the sport.

Python implementation

Naïve Bayes Classifier – Python implementation

Training Data:

- Weather: ['Sunny', 'Overcast', 'Rainy', 'Sunny']
- Play: ['Yes', 'Yes', 'No', 'No’]

- Calculate the player should play or not given the weather condition “sunny”
- Check P(yes|sunny) > p(no|sunny) ?

Naïve Bayes Classifier – Python implementation

Dictionaries: to store conditional probabilities, initialize with zeros
P(sunny|yes), p(overcast|yes), p(Rainy|yes) are represented as
prob_given_yes = {'Sunny': 0, 'Overcast': 0, 'Rainy': 0}

P(sunny|no), p(overcast|no), p(Rainy|no) are represented as
prob_given_no = {'Sunny': 0, 'Overcast': 0, 'Rainy': 0}

predict(weather): function to check the comparison based on argument :weather = sunny
Check P(yes|sunny) > p(no|sunny)

Naïve Bayes Classifier – Python implementation

```
#Training data
data = [
 ('Sunny', 'Yes'),
  ('Overcast', 'Yes’),
   ('Rainy', 'No’),
   ('Sunny', 'No’)]
#  Count how many times the class is Yes
yes_count = 0
#  Count how many times the class is No
no_count = 0
# Initialize dictionaries to store conditional probabilities
prob_given_yes = {'Sunny': 0, 'Overcast': 0, 'Rainy': 0}
prob_given_no = {'Sunny': 0, 'Overcast': 0, 'Rainy': 0}
```

These dictionaries will eventually store:
prob_given_yes P(Sunny | Yes) P(Overcast | Yes) P(Rainy | Yes)
prob_given_no P(Sunny | No) P(Overcast | No) P(Rainy | No)
But **at this stage they contain counts, not probabilities**.

Naïve Bayes Classifier – Python implementation

```
# Count the occurrences of 'Yes' and 'No' labels in the training data
for weather, label in data:
        if label == 'Yes':
                yes_count += 1
         elif label == 'No':
                 no_count += 1
print("yes_count=", yes_count)
print("no_count=", no_count)
# Calculate the probabilities of 'Yes' and 'No' labels
total_instances = len(data)
#Calculate prior probabilities
prob_yes = yes_count / total_instances
prob_no = no_count / total_instances

print("prob_yes=", prob_yes)
print("prob_no=", prob_no)
```

Naïve Bayes Classifier – Python implementation

```
# Calculate conditional probabilities for each weather condition given each label
for weather, label in data:
        if label == 'Yes':
                prob_given_yes[weather] += 1
        elif label == 'No':
                prob_given_no[weather] += 1
print("weather=", weather, "prob_given_yes=", prob_given_yes)
print("weather=", weather, "prob_given_no=", prob_given_no)
print("Final prob_given_yes", prob_given_yes)
print("Final prob_given_no", prob_given_no)
# Convert counts into conditional probabilities
for weather in prob_given_yes:
        prob_given_yes[weather] /= yes_count
for weather in prob_given_no:
        prob_given_no[weather] /= no_count
```

Therefore:P(Sunny | Yes) = 1/2 = 0.5 P(Overcast | Yes) = 1/2 = 0.5 P(Rainy | Yes) = 0/2 = 0

Naïve Bayes Classifier – Python implementation

```
# Define a function to predict the label based on weather
def predict(weather):
        prob_yes_given_weather = prob_yes * prob_given_yes[weather]
        prob_no_given_weather = prob_no * prob_given_no[weather]
        if prob_yes_given_weather > prob_no_given_weather:
                return 'Yes’
        else:
                return 'No’
# Test the classifier with a new input
new_input = 'Sunny’
predicted_result = predict(new_input)
print(f"Predicted label for {new_input}: {predicted_result}")
```

The function calculates:Yes score = P(Yes) × P(Sunny | Yes) = 0.5 × 0.5 = 0.25
No score = P(No) × P(Sunny | No) = 0.5 × 0.5 = 0.25
Since both are equal, this particular program chooses No because of the else

Example

Sl. No.	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

Using the given data, we have to identify the species of an entity with the following attributes.

X={Color=Green, Legs=2, Height=Tall, Smelly=No}

To implement a Naive Bayes classifier, we perform three steps.

1.First, we calculate the probability of each class label in the training dataset.

2.Next, we calculate the conditional probability of each attribute of the training data for each class label given in the training data.

3.Finally, we use the Bayes theorem and the calculated probabilities to predict class labels for new data points. For this, we will calculate the probability of the new data point belonging to each class. The class with which we get the maximum probability is assigned to the new data point.

To predict the class label for the above attribute set, we will first calculate the probability of the species being M or H in total.

P(Species=M)=4/8=0.5

P(Species=H)=4/8=0.5

Next, we will calculate the conditional probability of each attribute value for each class label.

P(Color=White/Species=M)=2/4=0.5

P(Color=White/Species=H)=¾=0.75

P(Color=Green/Species=M)=2/4=0.5

P(Color=Green/Species=H)=¼=0.25

P(Legs=2/Species=M)=1/4=0.25

P(Legs=2/Species=H)=4/4=1

P(Legs=3/Species=M)=3/4=0.75

P(Legs=3/Species=H)=0/4=0

P(Height=Tall/Species=M)=3/4=0.75

P(Height=Tall/Species=H)=2/4=0.5

P(Height=Short/Species=M)=1/4=0.25

P(Height=Short/Species=H)=2/4=0.5

P(Smelly=Yes/Species=M)=3/4=0.75

P(Smelly=Yes/Species=H)=1/4=0.25

P(Smelly=No/Species=M)=1/4=0.25

P(Smelly=No/Species=H)=3/4=0.75

We can tabulate the above calculations in the tables for better visualization.

The conditional probability table for the Color attribute is as follows.

Color	M	H
White	0.5	0.75
Green	0.5	0.25

Conditional Probabilities for Color Attribute

The conditional probability table for the Legs attribute is as follows.

Legs	M	H
2	0.25	1
3	0.75	0

Conditional Probabilities for Legs Attribute

The conditional probability table for the Height attribute is as follows.

Height	M	H
Tall	0.75	0.5
Short	0.25	0.5

Conditional Probabilities for Height Attribute

The conditional probability table for the Smelly attribute is as follows.

Smelly	M	H
Yes	0.75	0.25
No	0.25	0.75

Conditional Probabilities for Smelly Attribute

Now that we have calculated the conditional probabilities, we will use them to calculate the probability of the new attribute set belonging to a single class.

Let us consider X= {Color=Green, Legs=2, Height=Tall, Smelly=No}.

Then, the probability of X belonging to Species M will be as follows.

P(M/X)=P(Species=M)*P(Color=Green/Species=M)*P(Legs=2/Species=M)*P(Height=Tall/Species=M)*P(Smelly=No/Species=M)
=0.5*0.5*0.25*0.75*0.25
=0.0117

Similarly, the probability of X belonging to Species H will be calculated as follows.

P(H/X)=P(Species=H)*P(Color=Green/Species=H)*P(Legs=2/Species=H)*P(Height=Tall/Species=H)*P(Smelly=No/Species=H)
=0.5*0.25*1*0.5*0.75
=0.0468

So, the probability of X belonging to Species M is 0.0117 and that to Species H is 0.0468. Hence, we will assign the entity X with attributes {Color=Green, Legs=2, Height=Tall, Smelly=No} to species H.

Example

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$$

naive Bayes Classifier:

P(Refund=Yes|No) = 3/7
P(Refund=No|No) = 4/7
P(Refund=Yes|Yes) = 0
P(Refund=No|Yes) = 1
P(Marital Status=Single|No) = 2/7
P(Marital Status=Divorced|No)=1/7
P(Marital Status=Married|No) = 4/7
P(Marital Status=Single|Yes) = 2/7
P(Marital Status=Divorced|Yes)=1/7
P(Marital Status=Married|Yes) = 0

For taxable income:
If class=No: sample mean=110
sample variance=2975
If class=Yes: sample mean=90
sample variance=25

- P(X|Class=No) = P(Refund=No|Class=No)
  × P(Married| Class=No)
  × P(Income=120K| Class=No)
  = 4/7 × 4/7 × 0.0072 = 0.0024

- P(X|Class=Yes) = P(Refund=No| Class=Yes)
  × P(Married| Class=Yes)
  × P(Income=120K| Class=Yes)
  = 1 × 0 × 1.2 × 10^{-9} = 0

Since P(X|No)P(No) > P(X|Yes)P(Yes)

Therefore P(No|X) > P(Yes|X)

=> Class = No

Example

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

A: attributes

M: mammals

N: non-mammals

$$P(A \mid M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A \mid N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A \mid M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A \mid N)P(N) = 0.004 \times \frac{13}{20} = 0.0027$$

P(A|M)P(M) > P(A|N)P(N)

=> Mammals

Given all the previous patients I've seen (below are their symptoms and diagnosis)...

chills	runny nose	headache	fever	flu?
Y	N	Mild	Y	N
Y	Y	No	N	Y
Y	N	Strong	Y	Y
N	Y	Mild	Y	Y
N	N	No	N	N
N	Y	Strong	Y	Y
N	Y	Strong	N	N
Y	Y	Mild	Y	Y

Do I believe that a patient with the following symptoms has the flu?

chills	runny nose	headache	fever	flu?
Y	N	Mild	Y	?

First, we compute all possible individual probabilities conditioned on the target attribute (flu).

P(Flu=Y)	0.625		P(Flu=N)	0.375
P(chills=Y|flu=Y)	0.6		P(chills=Y|flu=N)	0.333
P(chills=N|flu=Y)	0.4		P(chills=N|flu=N)	0.666
P(runny nose=Y|flu=Y)	0.8		P(runny nose=Y|flu=N)	0.333
P(runny nose=N|flu=Y)	0.2		P(runny nose=N|flu=N)	0.666
P(headache=Mild|flu=Y)	0.4		P(headache=Mild|flu=N)	0.333
P(headache=No|flu=Y)	0.2		P(headache=No|flu=N)	0.3333
P(headache=Strong|flu=Y)	0.4		P(headache=Strong|flu=N)	0.333
P(fever=Y|flu=Y)	0.8		P(fever=Y|flu=N)	0.333
P(fever=N|flu=Y)	0.2		P(fever=N|flu=N)	0.666

And then decide:

$$argmax P(flu = Y)P(chills = Y|flu = Y)P(runny\ nose = N|flu = Y)P(headache = Mild|flu = Y)P(fever = N|flu = Y)$$
$$= 0.006$$

vs.

$$argmax P(flu = N)P(chills = Y|flu = N)P(runny\ nose = N|flu = N)P(headache = Mild|flu = N)P(fever = N|flu = N)$$
$$= 0.0185$$

Ques:) For the given dataset, Apply Naive-Baye's algorithm and Predict the outcome for a Car = {Red, Domestic, SUV}.

Color	Type	Origin	Stolen
Red	Sports	Domestic	Yes
Red	Sports	Domestic	NO
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	NO
Yellow	Sports	Imported	Yes
Yellow	SUV	Imported	NO
Yellow	SUV	Imported	Yes
Yellow	SUV	Domestic	NO
Red	SUV	Imported	NO
Red	Sports	Imported	Yes

0:08 / 11:13

Posterior $P(A|B) = \frac{P(B|A).P(A)}{P(B)}$

Prob. of B when A is True (Likelihood) → Proposition

Prob (A) when B is true

→ evidence

Prob (A) when B is true → evidence

$X = [Red, Domestic, SUV] = P(X|Yes) = \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} = \frac{6}{125} = 0.024$

i) $P(Red|Yes) = \frac{P(Yes|Red).P(Red)}{P(Yes)} = \frac{\frac{3}{5} \cdot \frac{5}{10}}{5/10} \cdot \frac{3}{5}$

ii) $P(Domestic|Yes) = 2/5$ iii) $P(SUV|Yes) = 1/5$

ii) $P(Domestic|Yes) = 2/5$ iii) $P(SUV|Yes) = 1/5$

$P(Red|No) = \frac{P(No|Red).P(Red)}{P(No)} = \frac{\frac{2}{5} \cdot \frac{5}{10}}{5/10} = 2/5$

$P(Domestic|No) = 3/5$, $P(SUV|No) = 2/5$ $= \frac{2 \times 3 \times 2}{5 \times 5 \times 5} = 0.07$

YouTube

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

$$p(Yes) = \frac{5}{10} = 0.5$$

$$p(No) = \frac{5}{10} = 0.5$$

Color	Yes	No
Red	3/5	2/5
Yellow	2/5	3/5

Type	Yes	No
Sports	4/5	2/5
SUV	1/5	3/5

Origin	Yes	No
Domestic	2/5	3/5
Imported	3/5	2/5

$$\boldsymbol{New\ Instance = (Red, SUV, Domestic) \rightarrow (Yes\ or\ No)}$$

$$P(Yes|New\ Instance) = p(Yes) * P(Color = Red|Yes) * P(Type = SUV|Yes) * P(Origin = Domestic|Yes)$$

$$P(Yes|New\ Instance) = \frac{5}{10} * \frac{3}{5} * \frac{1}{5} * \frac{2}{5} = \frac{3}{125} = 0.024$$

$$P(No|New\ Instance) = p(No) * P(Color = Red|No) * P(Type = SUV|No) * P(Origin = Domestic|No)$$

$$P(No|New\ Instance) = \frac{5}{10} * \frac{2}{5} * \frac{3}{5} * \frac{3}{5} = \frac{9}{125} = 0.072$$

$$P(No|New\ Instance) > P(Yes|New\ Instance)$$

MORE VIDEOS

Naïve Bayes Classifier

- If one of the conditional probability is zero, then the entire expression becomes zero
- Probability estimation:

$$\text{Original}: P(A_i \mid C) = \frac{N_{ic}}{N_c}$$

$$\text{Laplace}: P(A_i \mid C) = \frac{N_{ic} + 1}{N_c + c}$$

$$\text{m-estimate}: P(A_i \mid C) = \frac{N_{ic} + mp}{N_c + m}$$

c: number of classes

p: prior probability

m: parameter

Working of Naïve Bayes Classifier –Text Classification

Text Classification

- Assigning subject categories, topics, or genres
- Spam detection
- Authorship identification
- Age/gender identification
- Language Identification
- Sentiment analysis

Text Classification Definition

- *Input*:
- a document d
- a fixed set of classes $C = \{c1, c2,\ldots, cJ\}$
- *Output*: a predicted class $c \in C$

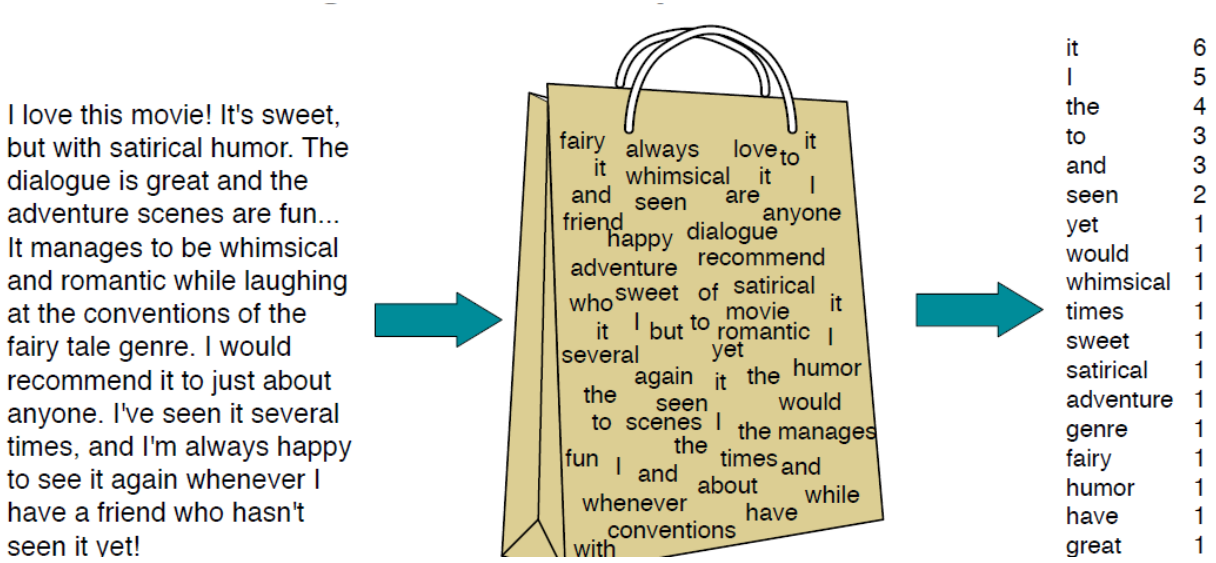
Text Classification: Supervised Machine Learning

- *Input:*
  - a document d
  - a fixed set of classes $C = \{c_1, c_2, \ldots, c_J\}$
  - A training set of m hand-labeled documents $(d_1,c_1),\ldots,(d_m,c_m)$
- *Output:*
  - a learned classifier $\gamma:d \rightarrow c$

Notation: use Y(d) = C to represent classifier
Y() is the classifier,
d is the document, and
c is the class we assigned to the document.
Ex: mark as spam button in Google

Text Classification: Naïve Bayes

- Relies on very simple representation of document
- Bag of words



Text Classification: Naïve Bayes Independence Assumptions

$$P(x_1, x_2, \ldots, x_n \mid c)$$

- **Bag of Words assumption**: Assume position doesn't matter
- **Conditional Independence**: Assume the feature probabilities $P(x_i \mid c_j)$ are independent given the class c.

$$P(x_1, \ldots, x_n \mid c) = P(x_1 \mid c) \bullet P(x_2 \mid c) \bullet P(x_3 \mid c) \bullet \ldots \bullet P(x_n \mid c)$$

Text Classification – Bayes Rule Applied to Documents and Classes

- For a document *d* and a class *c*

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

Naïve Bayes Classifier - Text Classification

- We can not feed in text directly to our classifier.
- Texts are huge, they have lots of words and various combinations
  - so **we use occurrences of words in a single text data**(i.e. term frequency (TF)) and
  - how many times that word comes across all text data (i.e. inverse document frequency-idf) and
  - then we can generate feature vectors.
- Feature vectors representing text contains the probabilities of appearance of the words of the text within the texts of a given category so that the algorithm can compute the likelihood of that text's belonging to the category.
- Once we are done finding likelihoods of various categories we find the category having maximum likelihood and there we have categorized our text

Working of Naïve Bayes Classifier –Text Classification 1

Naïve Bayes Classifier - Text Classification

Predict the class of “Please buy GPU from our store”

DOCUMENT ID	CONTENT	CLASS
1	Nvidia GPU is the best in the world.	computer graphics
2	Nvidia is giving tough competition to AMD.	computer graphics
3	We were running our application with GTX 1050(High end GPU) still it didn't work then we realized the problem was with the OS.	computer graphics
4	GPU, Ganpat Pandey University, is located in Maharashtra.	not computer graphics
5	Please buy GPU from our store.	?

Naïve Bayes Classifier - Text Classification

- Predict the class of “Please buy GPU from our store” by using Naive Bayes by considering the probability of important words

- Important words for computer graphics from the data can be
- {Nvidia, GPU, AMD, GTX 1050}

- All words will have feature vectors of representing their number of occurrences in each record

- Nvidia -> [1; 1; 0; 0]

Naïve Bayes Classifier - Text Classification

- For a document *d* and a class *c*

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

c= computer graphics
d = GPU
Calculate P(computer graphics | GPU)

Naïve Bayes Classifier - Text Classification

- P(computer graphics | GPU) = P(GPU | computer graphics) * (P(computer graphics) / P(GPU))
- P(computer graphics) = Number of records having class as computer graphics / Number of total records = 3/4 = 0.75
- P(GPU) = Number of records having GPU / Total number of records = 3/4 = 0.75
- P(GPU | computer graphics) = Number of records having computer graphics with GPU in it / Total number of computer graphics records = 2 /3 = 0.66
- P(computer graphics | GPU) = 0.66 * (0.75/0.75) = 0.66
- This is greater than 0.5 so we can predict that this text data will be belonging to computer graphics.

Note: Here, we have considered only one word GPU that is important in our test data

- But in real data consider feature vectors of all words and then compute the probability for the class based on the occurrences of all the non primitive words

Naïve Bayes Classifier - Text Classification 2

Naïve Bayes Classifier - Text Classification 2

- Build a classifier that says whether a text is about sports or not
- Training data has 5 sentences:

Text	Tag
"A great game"	Sports
"The election was over"	Not sports
"Very clean match"	Sports
"A clean but forgettable game"	Sports
"It was a close election"	Not sports

- Which tag does the sentence “A very close game” belong to?

Naïve Bayes Classifier - Text Classification 2

- Calculate the probability that the sentence "A very close game" is Sports
- Also calculate the probability that it is Not Sports.
- Take the largest one.
- Mathematically,
- P (Sports | a very close game) — the probability that the tag of a sentence is Sports given that the sentence is “A very close game”.

Naïve Bayes Classifier - Text Classification

- Feature Engineering
- The first thing when creating a machine learning model is to decide what to use as features.
- Features are information from the text to feed to the algorithm.
- Ex1: if we were doing classification on health, features could be a person's height, weight, gender, etc.
- We would exclude things that maybe are known but are not useful to the model, like a person's name or favorite color.

- Here, we have text which is to be converted into numeric features (numbers)

Naïve Bayes Classifier - Text Classification

- Use word frequencies
- So ignore word order and sentence construction, treating every document as a set of the words it contains.
- Features will be the counts of each of these words
- Note: to find out which tag has a bigger probability, we can discard the divisor —which is the same for both tags

- For a document *d* and a class *c*

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

c= sports
d = a very close game
Calculate P(sports| a very close game)

$$P(sports|a\ very\ close\ game) = \frac{P(a\ very\ close\ game|sports) \times P(sports)}{P(a\ very\ close\ game)}$$

Naïve Bayes Classifier - Text Classification

So, calculate the following:

$$P(a\,very\,close\,game|Sports) \times P(Sports)$$

$$P(a\,very\,close\,game|Not\,Sports) \times P(Not\,Sports)$$

We could actually calculate these probabilities: count how many times the sentence “A very close game” appears in the Sports tag, divide it by the total, and obtain P (a very close game | Sports).

Difficulty: “A very close game” does not appear in our training data.
so this probability is zero.

Naïve Bayes Classifier - Text Classification

- What is naïve here?
- Assume that every word in a sentence is independent of each other
- In practice, this is rarely the case. This is why the name **Naïve** Bayes
- So, we are no longer looking at entire sentences, but rather at individual words.
- We know that individual words actually show up several times in our training data
- So "this was a fun party" is the same as "this party was fun" and "party fun was this".

Naïve Bayes Classifier - Text Classification

- Calculating Probabilities

$$P(a\,very\,close\,game) = P(a) \times P(very) \times P(close) \times P(game)$$

- Appying to Bayes theorem:

$$P(a\,very\,close\,game|Sports) = P(a|Sports) \times P(very|Sports) \times P(close|Sports) \times P(game|Sports)$$

- Calculate the a priori probability of each tag:
- for a given sentence in our training data, the probability that it is Sports P (Sports) =3/5
- P (Not Sports) = 2/5
- calculate P (game | Sports): counting how many times the word “game” appears in Sports texts (2) divided by the total number of words in sports (11).
- P (game | Sports) = 2/11

Naïve Bayes Classifier - Text Classification

- Calculating Probabilities
- “close” does not appear in any Sports text.
- So P (close | Sports) = 0

$$P(a\ very\ close\ game|Sports) = P(a|Sports) \times P(very|Sports) \times P(close|Sports) \times P(game|Sports)$$

$$P(a|Sports) \times P(very|Sports) \times 0 \times P(game|Sports)$$

- P(very| not Sports) = 0
- P (game | not sports) =0
- We end up with zero.
- Problem with MLE(Max Likelihood estimation)
- Solution: Laplace smoothing (add-1)

Naïve Bayes Classifier - Text Classification

$$\hat{P}(w_i \mid c_j) = \frac{count(w_i, c_j)}{\sum_{w \in V} count(w, c_j)}$$

fraction of times word w_i appears among all words in documents of topic c_j

Laplace (add-1) smoothing for Naïve Bayes

$$\hat{P}(w_i \mid c) = \frac{count(w_i, c) + 1}{\sum_{w \in V} \left(count(w, c) + 1 \right)}$$

$$= \frac{count(w_i, c) + 1}{\left(\sum_{w \in V} count(w, c) \right) + |V|}$$

Naïve Bayes Classifier - Text Classification

- Calculating Probabilities : Laplace smoothing
- Add 1 to every count so there is no zero.
- To balance this, add the number of possible words to the divisor, so the division will never be greater than 1.
- Here the possible words are ['a', 'great', 'game', 'the', 'election', 'was', 'over', 'very', 'clean , 'match', 'but', 'forgettable', 'it', ', 'close'].
- Since the number of unique words is 14, applying smoothing we get that
- Note: Actual value without smoothing P (game | Sports) = 2/11.
- Applying smoothing: V= 14

$$P(game|sports) = \frac{2+1}{11+14}$$

Naïve Bayes Classifier - Text Classification

- Calculating Probabilities –classifier gives “A very close game” the Sports tag

Word	P (word | Sports)	P (word | Not Sports)
a	(2 + 1) ÷ (11 + 14)	(1 + 1) ÷ (9 + 14)
very	(1 + 1) ÷ (11 + 14)	(0 + 1) ÷ (9 + 14)
close	(0 + 1) ÷ (11 + 14)	(1 + 1) ÷ (9 + 14)
game	(2 + 1) ÷ (11 + 14)	(0 + 1) ÷ (9 + 14)

Text	Tag
"A great game"	Sports
"The election was over"	Not sports
"Very clean match"	Sports
"A clean but forgettable game"	Sports
"It was a close election"	Not sports

$$P(a|Sports) \times P(very|Sports) \times P(close|Sports) \times P(game|Sports) \times P(Sports)$$
$$= 2.76 \times 10^{-5}$$
$$= 0.0000276$$

$$\mathrm{P(a}\,|Not\ Sports) \times P(very|Not\ Sports) \times P(close|Not\ Sports) \times P(game|Not\ Sports) \times P(Not\ Sports)$$
$$= 0.572 \times 10^{-5}$$
$$= 0.00000572$$

Naïve Bayes Classifier - Text Classification

- Unknown words
- Words that occur in test data but not in our vocabulary in any training document in any class - unknown words
- Solution for such unknown words is to ignore them—remove them from the test document and not include any probability
- With naive Bayes text classification, ignore any word that never occurred in the training document.
- Ex: “A clean tough game”. Here “tough” does not appear in the training data
- So change the test data to “A clean game”
- What about words that appear in some classes but not others ?

Naïve Bayes Classifier - Text Classification

- When to apply Add-one (Laplace) Smoothing ?
- Retain words that appear in some classes but not others and use Smoothing
- How to find the size of Vocabulary V?
- Vocabulary V is the number of unique words in our training data
- Vocabulary V consists of the union of all the word types in all classes, not just the words in one class

Naïve Bayes Classifier - Text Classification

- Advanced techniques: to improve this basic model of Naive Bayes

- Removing stop words. These are common words that do not add anything to the classification, such as a, able, either, else, ever and so on. So “The election was over” would be “election over” and “a very close game” would be “very close game”.

- Lemmatizing words. This is grouping together different inflections of the same word. So “election”, “elections”, “elected”, and so on would be grouped together and counted as more appearances of the same word.

- Using n-grams. Instead of counting single words, we could count sequences of words, like “clean match” and “close election”.

- Using TF-IDF. Instead of counting frequency we could do more advanced like penalizing words that appear frequently in most of the texts.

Gaussian Naive Bayes —

•In a Gaussian Naive Bayes, the predictors take a continuous value assuming that it has been sampled from a Gaussian Distribution.

•It is also called a Normal Distribution.

Patient	Blood Pressure	Disease
1	120	No
2	125	No
3	130	No
4	140	Yes
5	145	Yes
6	150	Yes

We want to predict the disease status of a new patient with:X=BP=142

We want to predict the disease status of a new patient with:X=BP=142

Step 1: Calculate mean for each class

For **Disease = No**:

$$BP = 120,125,130$$

Mean:

$$\mu_{No} = \frac{120 + 125 + 130}{3} = 125$$

For **Disease = Yes**:

$$BP = 140,145,150$$

Mean:

μYes=3140+145+150=145

Step 2: Calculate standard deviation

For simplicity, using population standard deviation:

For No:

$$\sigma_{No} \approx 4.08$$

For Yes:

$$\sigma_{Yes} \approx 4.08$$

Step 3: Calculate Gaussian probability for BP = 142

Gaussian probability density is:

$$P(x \mid C) = \frac{1}{\sqrt{2\pi}\sigma_C} e^{-\frac{(x-\mu_C)^2}{2\sigma_C^2}}$$

For **Disease = No**:

P(142|No)

Since 142 is far from the mean 125, the probability density will be **very small**.

For **Disease = Yes**:

P(142|Yes)

Important distinction

Consider these two situations:

Case 1 — Categorical feature

Blood pressure category:

Low, Normal, High

Use **Categorical Naive Bayes** and possibly Laplace smoothing.

Case 2 — Numerical feature

Actual blood pressure:

120, 125, 130, 140, 145, 150

Use **Gaussian Naive Bayes**.

So even though the training values are a finite/discrete set of numbers, **BP is a continuous numerical variable conceptually**, and Gaussian NB models its distribution.

Small training dataset

Assume we have **10 previously classified emails**:

Email	Content	Class
E1	Win free prize	Spam
E2	Win free money	Spam
E3	Free prize now	Spam
E4	Win lottery prize	Spam
E5	Free offer	Spam
E6	Meeting at office	Ham
E7	Project meeting today	Ham
E8	Office meeting today	Ham
E9	Project discussion	Ham
E10	Meeting tomorrow	Ham

Step 1 — Calculate Prior Probability

Prior probability means:

How likely is each class before looking at the words in the new email?

Spam

$$P(Spam) = \frac{5}{10} = 0.5$$

Ham

$$P(Ham) = \frac{5}{10} = 0.5$$

Therefore:

Class	Prior
Spam	0.5
Ham	0.5

Step 2 — Calculate Likelihood

Now we look at the words in:

Win free prize

The words are:

- win
- free
- prize

We need to calculate:

$$P(Win \mid Spam)$$
$$P(Free \mid Spam)$$
$$P(Prize \mid Spam)$$

and similarly for Ham.

This is called the **likelihood**.

Count the words in Spam emails

Look only at the 5 Spam emails:

1. Win free prize
2. Win free money
3. Free prize now
4. Win lottery prize
5. Free offer

Let's count:

Word	Number of Spam emails containing word
win	3
free	4
prize	3

Therefore:

$$P(win \mid Spam) = \frac{3}{5} = 0.6$$
$$P(free \mid Spam) = \frac{4}{5} = 0.8$$
$$P(prize \mid Spam) = \frac{3}{5} = 0.6$$

Calculate likelihood for Spam

Naïve Bayes makes the **naïve assumption** that the words can be treated as conditionally independent given the class.

Therefore:

$$P(W, i, nFreePrize \mid Spam)$$

is approximated as:

$$P(Win \mid Spam) \times P(Free \mid Spam) \times P(Prize \mid Spam)$$

Substitute the values:

$$= 0.6 \times 0.8 \times 0.6$$
$$= \boxed{0.288}$$

So the likelihood of seeing these words given that the email is Spam is:

$$\boxed{P(Words \mid Spam) = 0.288}$$

Step 3 — Calculate Posterior Probability

Now we combine:

Prior × Likelihood

For Spam:

$$P(Spam) \times P(Words \mid Spam)$$
$$= 0.5 \times 0.288$$

$$= \square 0.144$$

For Ham:

$$P(Ham) \times P(Words|Ham)$$
$$= 0.5 \times 0.0000823$$
$$= \square 0.00004115$$

Calculate likelihood for Ham

Now look at the 5 Ham emails:

1. Meeting at office
2. Project meeting today
3. Office meeting today
4. Project discussion
5. Meeting tomorrow

Notice that:

- **win** occurs in 0 Ham emails
- **free** occurs in 0 Ham emails
- **prize** occurs in 0 Ham emails

If we directly calculate:

$$P(win \mid Ham) = 0$$
$$P(free \mid Ham) = 0$$
$$P(prize \mid Ham) = 0$$

Then:

$$P(Words \mid Ham) = 0 \times 0 \times 0 = 0$$

This would make the **probability of Ham zero**.
In real Naïve Bayes, **we solve this problem using Laplace smoothing.**

Laplace smoothing — simple idea

- Instead of allowing a word probability to become zero, we add 1 to each count.
- For example:
- P(win|Ham)=(0+1)/(5+V)

- where V is the number of unique words in our vocabulary.
- Suppose our vocabulary contains 18 unique words.

Then: $P(win \mid Ham) = \frac{1}{5+18} = \frac{1}{23} \approx 0.0435$

The complete flow is:

```
Training Emails
       ↓
Count Spam and Ham emails
       ↓
Calculate PRIOR
       ↓
Calculate how frequently each word
occurs in Spam and Ham
       ↓
Calculate LIKELIHOOD
       ↓
For a new email:
"Win free prize"
       ↓
Prior × Likelihood
       ↓
Calculate POSTERIOR probability
       ↓
Compare:
P(Spam | words)
       vs
P(Ham | words)
       ↓
Higher probability wins
       ↓
        SPAM
```